

Alan Zhang

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EXPERIENCE

Software Engineer

May 2025 – Current

Toll Brothers (Fortune 500, \$10B+ revenue, nationwide real estate platform)

Fort Washington, PA

- Shipped an LLM-powered plot plan QA platform for ESE Consultants (Toll Brothers subsidiary), assisting manual land survey reviews with Claude-powered compliance analysis, saving **2 hours per submission** across **5-10 daily nationwide submissions** with an admin-configurable prompt layer and real-time usage analytics.
- Designed an LLM-powered code review pipeline (Claude Sonnet) integrated into CI/CD - parsing unified diffs, resolving full context, and posting inline comments, **reducing average review time by 30 minutes/PR**.
- Built a Node.js CLI pipeline that ingests PDF batches and orchestrates uploads across the CMS and Workfront APIs — replacing a week-long manual workflow across several developers, saving **100+ hours per batch**.
- Launched a site-wide [HomePrint “Print View” system](#) serving **thousands of monthly** home-site visitors, generating community-specific brochure views with dynamic listing data, QR codes, and filter/sort functionality.

PROJECTS

YSC Lunch Soccer | *Next.js, React Native, Express.js, Supabase, PostgreSQL, Redis, BullMQ* [Live Site](#) | [GitHub](#)

- Replaced a group-chat coordination workflow for a **200-member** soccer community by building a full-stack session management platform with Stripe-integrated payments, attendance tracking, and role-based admin tools.
- Implemented real-time attendance updates via Supabase Realtime (Postgres CDC over WebSocket) with Redis-backed caching and endpoint-tiered rate limiting deployed across Vercel and Render.
- Decoupled notification delivery from the request path by building a BullMQ + Redis job queue — Twilio SMS and transactional emails processed asynchronously by a dedicated worker with **3-attempt exponential backoff**, eliminating blocking I/O and adding fault tolerance to all outbound notifications.
- Engineered a skill-based team balancing algorithm that groups players into **6 skill tiers**, randomizes players within each tier, and distributes them across dynamically sized teams using a snake draft, eliminating **~10 minutes** of manual session setup while producing more balanced team compositions.

ApplyLog | *Chrome Extension (MV3), TypeScript, Next.js, Supabase, Gmail API, OAuth 2.0* [Live Site](#) | [GitHub](#)

- Built a Chrome Extension (MV3) that scrapes job postings across Greenhouse, Lever, Ashby, and Workday behind a single dispatch interface, using URL-pattern matching to detect ATS providers and filter out non-posting pages (confirmation screens, login walls), syncing **500+ applications** to Supabase.
- Implemented PKCE OAuth 2.0 via `chrome.identity` — generating SHA-256 code challenges, exchanging authorization codes against Supabase’s token endpoint, and deduplicating concurrent token refresh calls on expiry to eliminate forced re-authentication.
- Designed an event-driven matching pipeline: a webhook fires on every scraped job, embeds the description with OpenAI, and computes cosine similarity against the user’s existing resume embedding, writing the match score back in a single keyed update.
- Versioned resume uploads on Supabase Storage under per-user-namespaced paths, pairing each PDF with its extracted text and embedding at a given version; served previews through short-lived signed URLs rather than public links to keep sensitive files access-controlled.

EDUCATION

University of Maryland – College Park

Aug. 2022 – May 2025

Bachelor of Science in Computer Science

College Park, MD

TECHNICAL SKILLS

Programming Languages: JavaScript, TypeScript, Python, Java

Front-end/Back-end/Mobile Technologies: React.js, React Native, Next.js, Node.js, HTML, CSS, SCSS, Tailwind CSS, GSAP, Flask, Express.js, GraphQL, REST APIs, Stripe, Twilio, Expo, Expo Router

Database & ORM Technologies: PostgreSQL, MySQL, MongoDB, Redis, DynamoDB, Prisma, SQL

Cloud & DevOps: AWS Amplify, Cognito, S3, CloudFront, Git, GitHub, GitLab, Bitbucket, Docker, Jira, Supabase